

Introducción a la Ciencia de datos con Python

Fenómeno / Problema

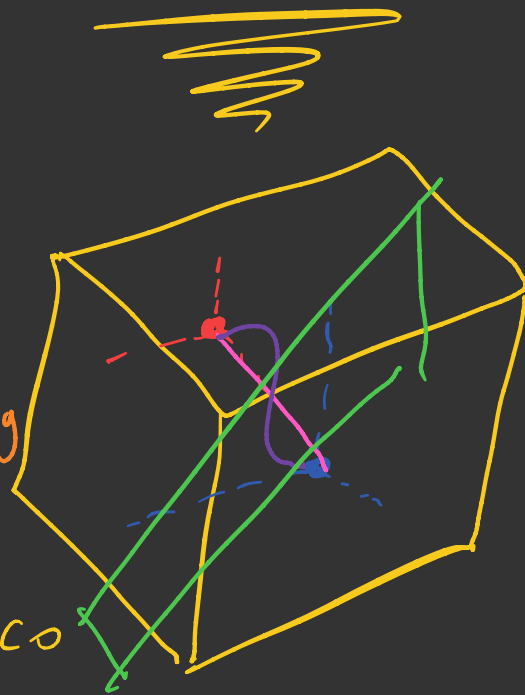
Datos

Numéricos
↑
Categoricos
↓
Texto

Embedding

Espacio Semántico

Espacio numérico



Conjunto de datos

Espacio de características

Tabla de Hechos

↳ características Columna

(Ejes de información)

↳ Observaciones Fila

(Punto de información)

Accidentes

	Edd	Sexo	Zona	
1	23	H	A	O ₁
2	35	M	B	O ₂
3	46	H	C	O ₃
	C ₁	C ₂	C ₃	

→ Edad

nombre | sexo | edad
└───┴───┴───┘
● persona

Espacio de características

- Numérica
- categórica
- Texto

Espacio numérico

c. Numéricas

- Edad
- Altura
- Salario
- Temperatura
- Peso

Estadística

- Media (Promedio)

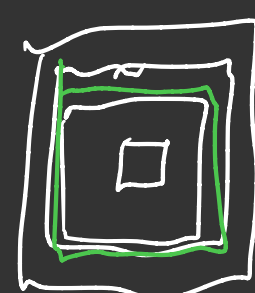
$$\bar{x}^{(1)} = \frac{\sum x_i^{(1)}}{n}$$



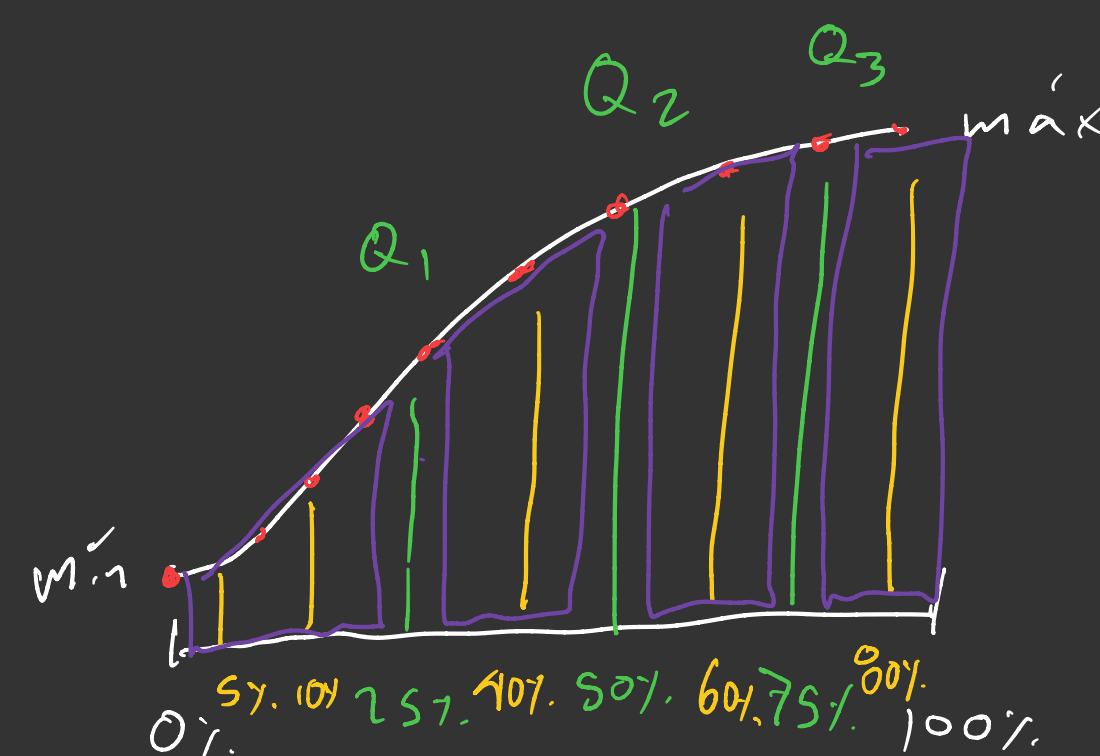
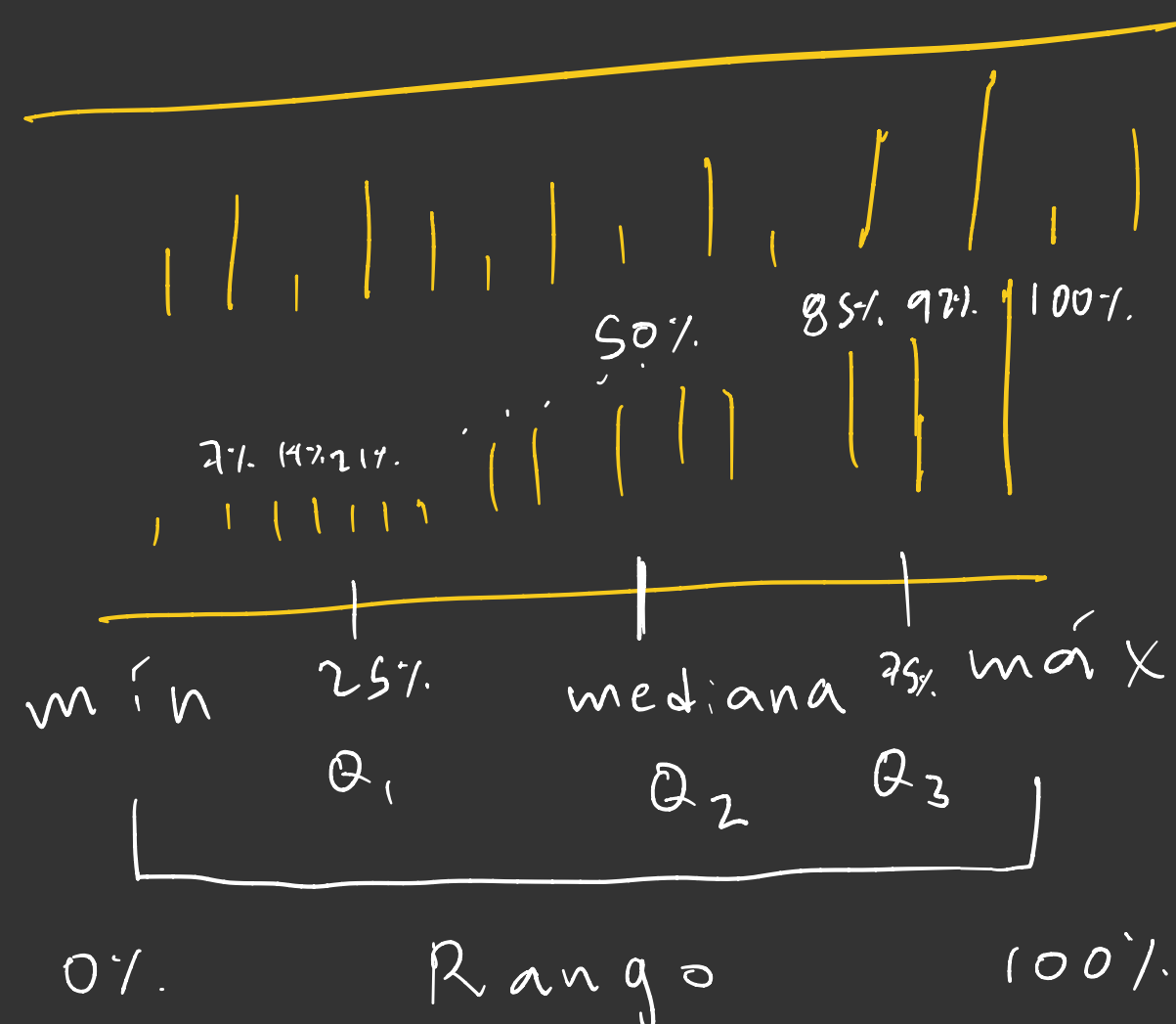
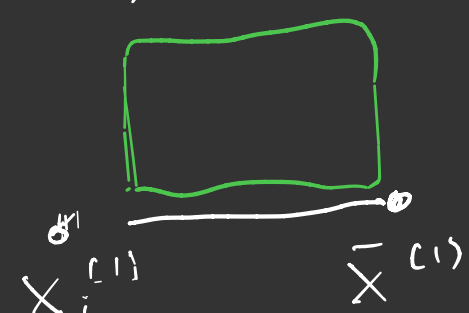
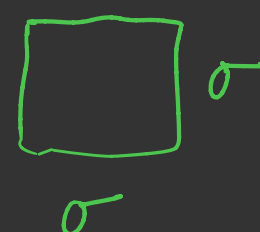
Desigual Proporción

- Varianza

$$\sigma_{(1)}^2 = \frac{\sum (x_i^{(1)} - \bar{x}^{(1)})^2}{n-1}$$



$$\sigma_{(1)} = \sqrt{\sigma_{(1)}^2}$$



	A	B
Q ₁	\$ 6,200	\$ 7,500
Q ₂	\$ 8,700	\$ 11,200
Q ₃	\$ 10,100	\$ 15,300

$$P(x_i^{(1)} < Q_1) = 0.25$$

$$P(x_i^{(1)} > Q_1) = 0.75$$

$$P(x_i^{(1)} < Q_2) = 0.5$$

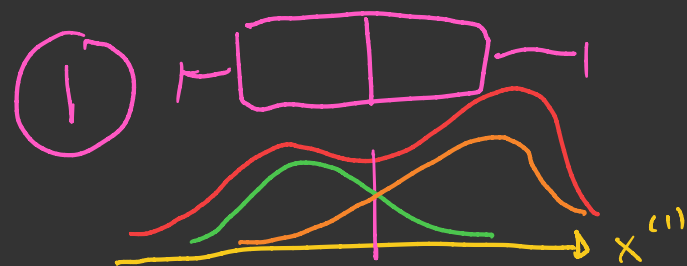
$$P(x_i^{(1)} > Q_2) = 0.5$$

$$P(x_i^{(1)} < Q_3) = 0.75$$

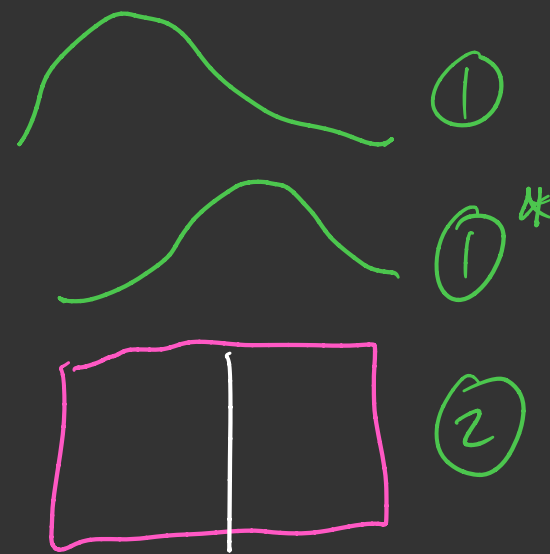
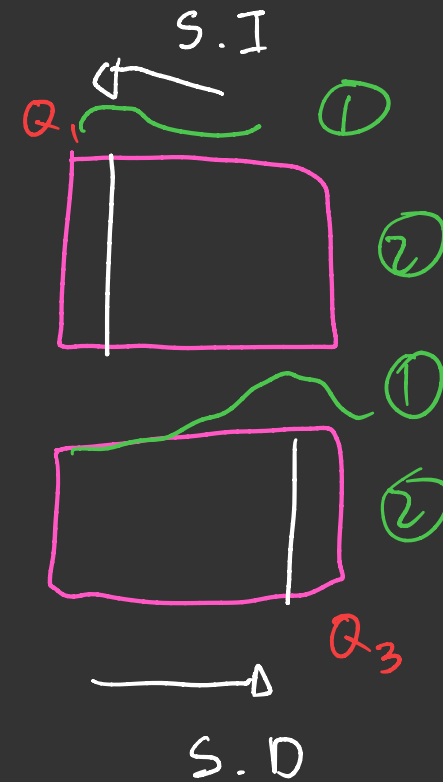
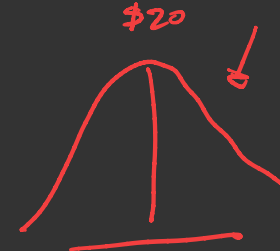
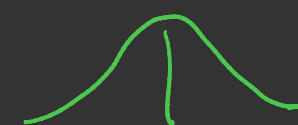
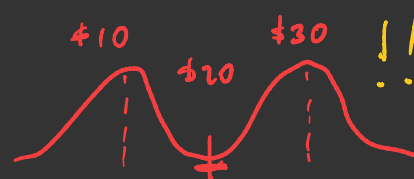
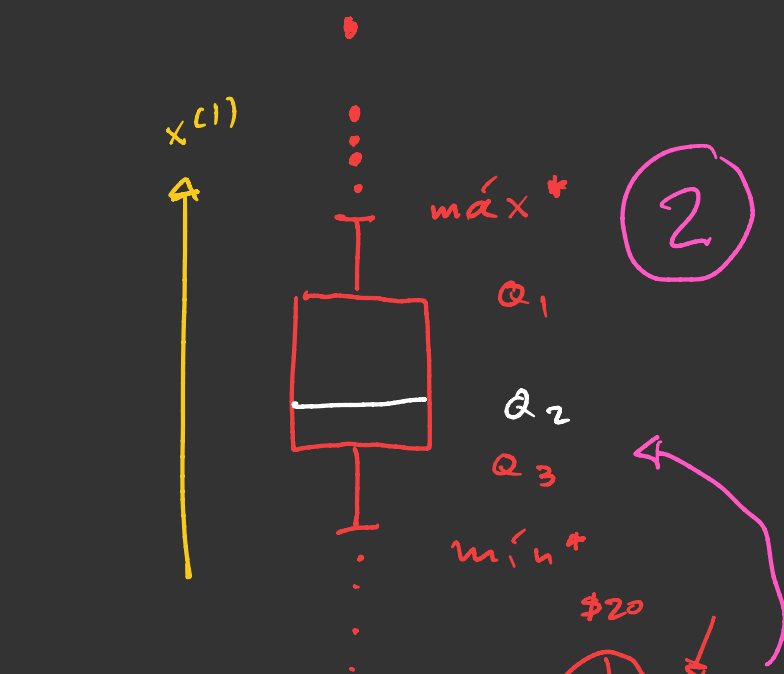
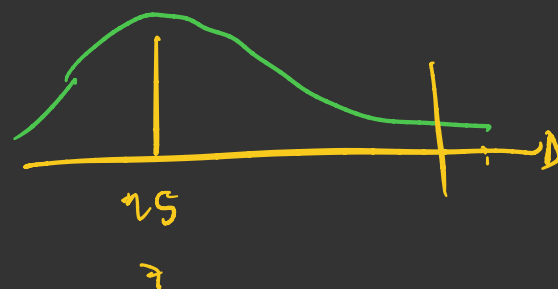
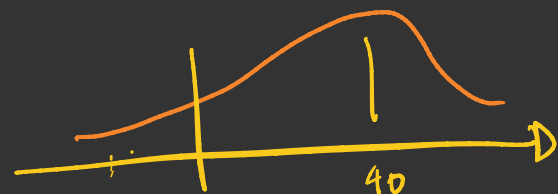
$$P(x_i^{(1)} > Q_3) = 0.25$$

$$P(Q_1 < x_i^{(1)} < Q_3) = 0.5$$

1- Eje numérico



No creer en la estadística*



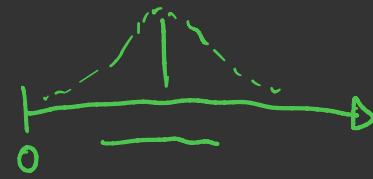
Transformación Box-Cox

del logaritmo

$$x^{(1)} \quad x^{(1)'} = \ln(x^{(1)})$$

$$\ln(1,000,000) = 12 \quad = \ln(1 + x^{(1)})$$

$$\ln(1,000,000,000) = 23 \quad = \ln(1 + \text{mín} + x^{(1)})$$

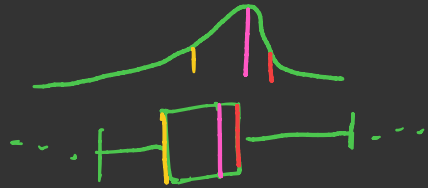


Visualización de datos

1-Eje

N Numérico

Densidad



C Categórico



2-Ejes

N Numérico - N Numérico



N Numérico - C Categórico



C Categórico - N Numérico

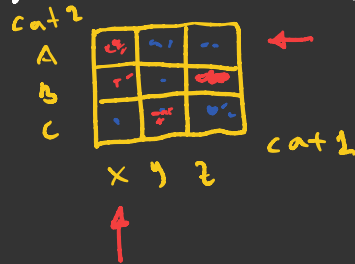


2 cat + 3 cat	
A 60	X 20
B 40	Y 30
	Z 30

6-cat	
Ax	30
Ay	20
Az	10
Bx	10
By	10
Bz	20

Categoría Cruzada

C Categórico - C Categórico



T. Mascota		C. Vacuna	
A	Perro	X	Si
B	Gato	Y	No
C	Otro	Z	Nosé